

Cloudflare Radar 2024 Year in Review

Data analysis report for technology executives and network infrastructure leaders

17% Global Traffic Growth and Starlink's 3.3x Surge

Global traffic up 17% YoY; satellite connectivity accelerates

17% YoY

Global Internet traffic growth in 2024

3.3×

Starlink traffic growth in 2024

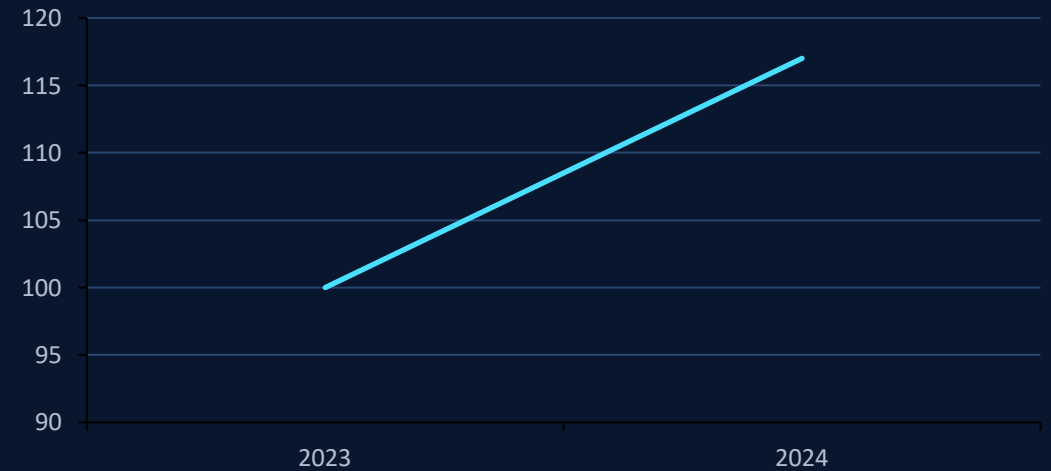
What it suggests

Trend normalized from mid-January to reduce post-holiday effects.

Starlink growth reflects expanding satellite Internet availability; scaled to peak observed volume.



Global Traffic (Index)



Index view (2023=100) illustrates the reported 17% YoY increase.

Google Leads Top 10 Services as Generative AI Rises

Ranked by Cloudflare 1.1.1.1 DNS resolver aggregates

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Context

Google held #1 for the second consecutive year.
Generative AI continued its rapid rise through 2024; “metaverse” momentum faded.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

PQC inflection after Chrome 124 default enablement (Apr 17, 2024)



2% → 13%

Post-quantum share of TLS 1.3 traffic (2024)

21%

HTTP/3 share of traffic

~12%

Go share of automated API requests

What leaders should note

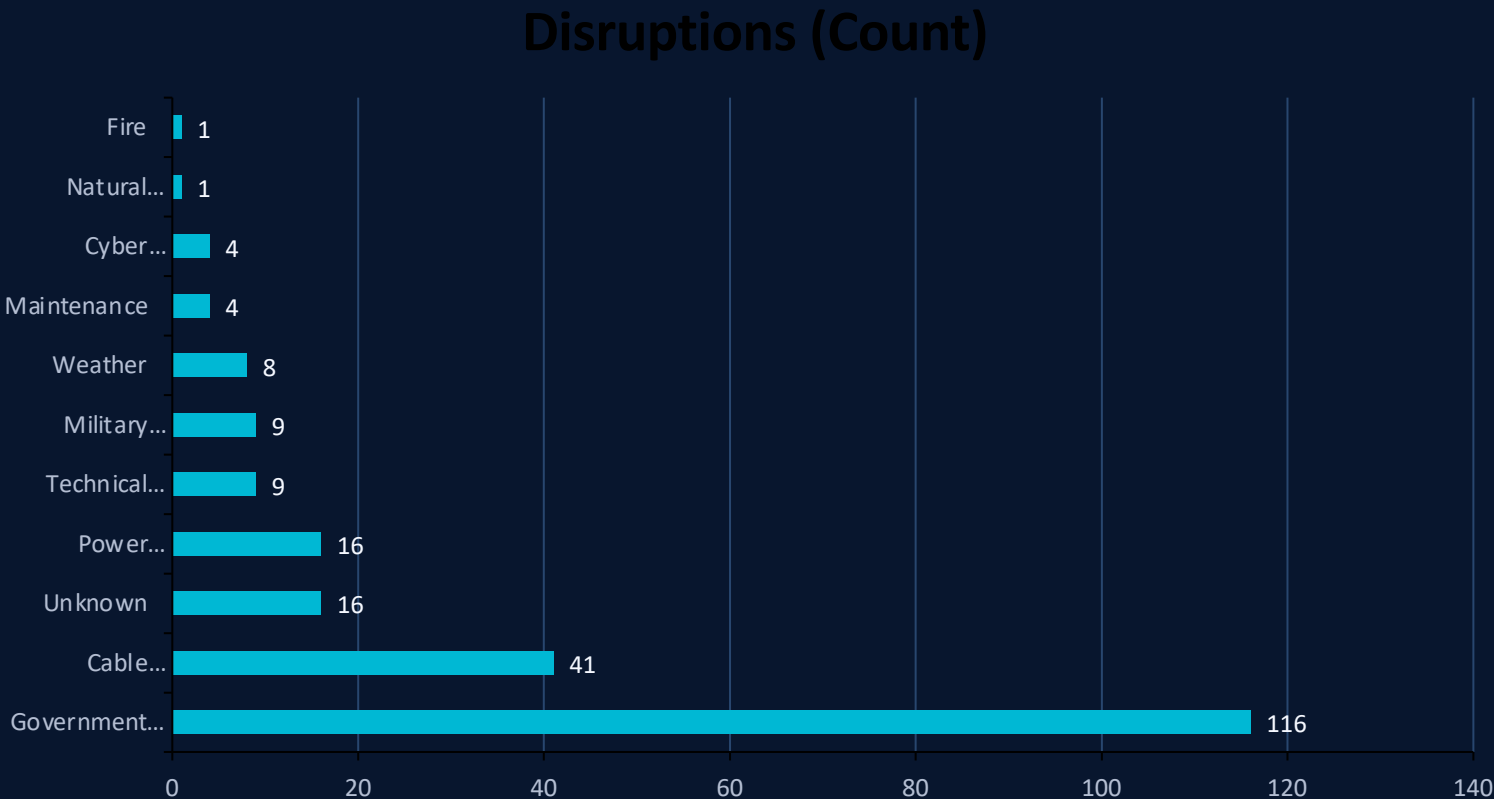
PQC surged sharply after April 17, 2024; stabilized around ~12–13%.

HTTP/3 reached 21% adoption (QUIC-based, encrypted-by-default).

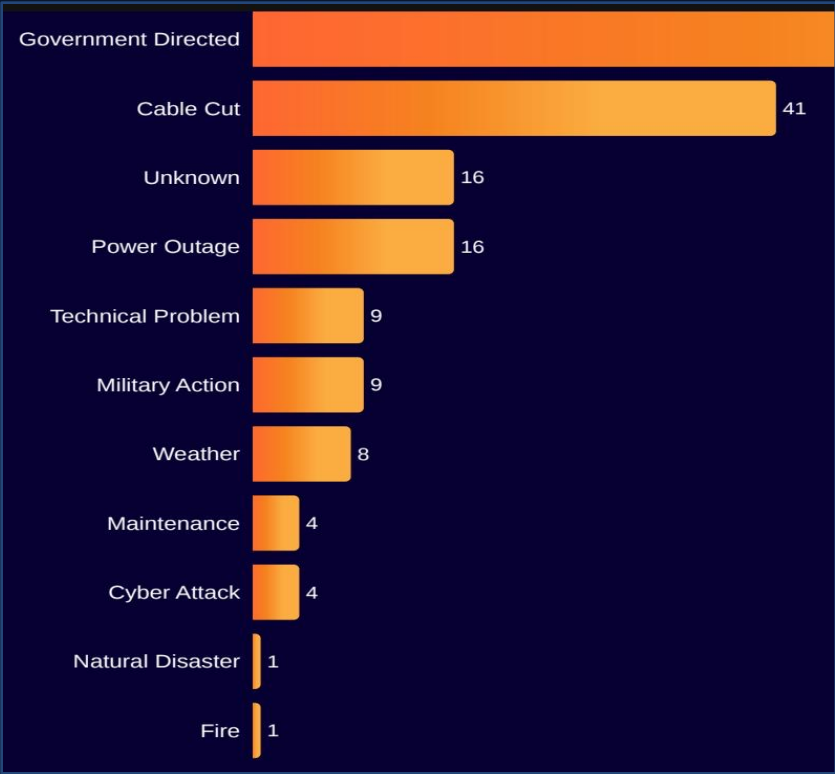
Go became the top API client language for automated API requests (~12%).

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare observed 225 major Internet disruptions globally in 2024



Note: Government-directed shutdowns are reported as 116+ in the source.



Implications

Government-directed shutdowns were the leading cause (116+).

Next highest: cable cuts (41) and power outages (16).

Plan resilience for policy-driven and physical-layer events.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security & email threat signals observed in 2024

6.5%

Share of total traffic mitigated

68%

Bot traffic from top 10 countries

4.3%

Emails that were malicious

Network security

Most-attacked sector: Gambling and Games.

Log4j (CVE-2021-44228) remains actively exploited years after disclosure.

RPKI valid routes increased by 6.4% (routing security progress).

Email security

Malicious email rate: 4.3% of emails processed.

43% of malicious emails contained a deceptive link.

Top malicious/spam TLD observed: .bar.

Takeaway: mitigation volume and long-tail vulnerabilities require continuous controls.

Five Takeaways Shaping the 2025 Internet Outlook

Strategic implications for network and security leaders

1

Plan capacity for continued demand: global traffic grew 17% YoY.

2

Prepare crypto agility: post-quantum TLS 1.3 rose from ~2% to ~13% after April 2024.

3

Harden for policy risk: government-directed shutdowns led 225 major disruptions (116+).

4

Assume persistent adversaries: 6.5% of traffic required mitigation; Log4j remains exploited.

5

Optimize for the developer shift: Go leads API client languages (~12%); HTTP/3 at 21% adoption.